Akula Saran Mani Kumar

Data Scientist

SUMMARY

AI/ML enthusiast with hands-on experience in Machine Learning, Deep Learning, Generative AI, and NLP through end-to-end projects. Skilled in Python, LangChain, OpenAI APIs, and Prompt Engineering for building intelligent and data-driven applications. Strong foundation in model development, problem solving, and AI solution deployment.

SKILLS

Gen AI
LLMs, Prompt Engineering, LangChain, OpenAI API, Hugging Face

Deep Learning
Neural Networks, ANN, CNN, RNN, LSTM, BiLSTM, TensorFlow, Keras, NLP, Model Evaluation

Machine Learning
Regression, Classification, Clustering, Cross Validation, Hyperparameter Tuning

Python
Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn

EDUCATION

B.E
Chaitanya Bharathi Institute Of Technology
2019 – 2023

CERTIFICATES

- Python | Machine Learning | EDA | Power BI | SQL — Certification at Innomatics Research Labs

ACHIEVEMENTS

1. 3rd Place in hackathon conducted by Grid dynamics

2. 5 star Coder in sql and python in Hackerrank

PROJECTS

1. LangChain Based Zomato Assistant
- Developed an **conversational Zomato chatbot** using **LangChain** and **OpenAI GPT-4o-mini** for personalized food and restaurant recommendations.
- Implemented **Conversation Memory** to retain user preferences and enable multi-turn conversations.
- Designed dynamic prompt templates and applied **Prompt Engineering** to improve response quality and personalization.
- Built an **LLM-based recommendation pipeline** supporting cuisine discovery, dietary preferences, and food assistance.

2. Fake news Detection using bidirectional lstm
- Developed a Fake News Detection system using Bidirectional LSTM and Word2Vec on **44K+ news articles** for binary text classification.
- Performed text preprocessing, tokenization, embedding generation, sequence padding, and feature engineering.
- Trained and optimized a deep learning pipeline with Early Stopping, achieving **99.83% test accuracy.**
- Deployed the model using Streamlit and Hugging Face for real-time fake news prediction.

3. Disease Prediction System
- Developed a **Disease Prediction System** using **BiLSTM** to classify diseases from **symptom text inputs** across disease categories.
- Built an **NLP pipeline** with text preprocessing, tokenization, sequence padding, and label encoding for accurate prediction
- Implemented Top-3 disease prediction with confidence scores, achieving **~87% validation accuracy** and improving prediction interpretability.
- Deployed the solution with **Streamlit** and integrated saved model artifacts (.keras, .pkl) for real-time inference.

4.Battery Life Prediction using ANN
- Developed an Artificial Neural Network (ANN) based predictive model to estimate **Battery Remaining Useful Life** using charging and discharge cycle data consisting of more than **1,00,000** datapoints.
- Performed **data preprocessing, feature engineering, and normalization** on battery parameters to improve prediction performance.